

A Theory of Economic Slack

Pascal Michailat

Draft version: August 2026

Draft URL: pascalmichailat.org/18/

20	TEN OPEN QUESTIONS	3
20.1	Why are unemployment fluctuations perennial?	3
20.2	What determines market price norms?	4
20.3	Why does the Beveridge curve shift?	5
20.4	Why is the product Okun curve steeper in the model than in the data? . .	5
20.5	Why is the monetary multiplier larger in the model than in the data? . .	5
20.6	Why is the fiscal multiplier smaller in the model than in the data? . . .	6
20.7	What is the welfare cost of unemployment?	6
20.8	What is the welfare cost of recruiting?	7
20.9	Can aggregate slack be inferred from labor market slack?	7
20.10	Does slack also influence longer-run fluctuations?	8
	BIBLIOGRAPHY	9

CHAPTER 20.

Ten open questions

In the previous chapter we reviewed the main takeaways from the book. My hope is that the book might help us understand markets and business cycles better. But obviously, there remains a number of unanswered questions, which hopefully will be answered in the future and further improve our understanding of markets and business cycles. This chapter lists ten open questions—both theoretical and empirical—that I think are particularly important.

20.1. Why are unemployment fluctuations perennial?

The first open question is why US unemployment fluctuates perennially—as shown in figure 2.1. Why don't we see long periods of stable unemployment?

In this book we saw how shocks to aggregate demand produce realistic fluctuations in unemployment; but it still remains to explain why these shocks seem to occur perennially. It seems that each recession has its own story; yet downturns keep occurring like clockwork after periods of expansion and confidence.

One possibility is that some of the events unleash the “animal spirits” that lie dormant otherwise, as Keynes (1936, chapter 12) proposed. It might be that after periods of stability, the animal spirits always awaken and produce bubbles and then crashes. This could explain the dot-com recession and Great Depression (stock market bubbles), the Great Recession (housing market bubble), and maybe the next recession (bubble around artificial intelligence and cryptocurrency). Speculative manias, followed by financial crashes,

have happened time and time again throughout history and around the world, driven by irrational exuberance from investors, and often exacerbated by fraudulent behavior.¹

More generally, animal spirits might be an irrational fervor that drives economic decisions of crowds of people, possibly ignited by news of certain events, and assuredly exacerbated by media coverage.² When animal spirits are triggered, they may lead to brash, large-scale reactions—consumers suddenly cutting spending and firms suddenly cutting hiring. The aggregate demand shock in the model is a simple way to formalize these swings in collective optimism.

In the book, the aggregate-demand shifter is exogenous. Another possibility is that business cycles are not driven by exogenous shocks at all (Beaudry, Galizia, and Portier 2020). Maybe business cycles are endogenous, not caused by shocks but due to circular or orbital dynamics around an equilibrium of the model. Diamond and Fudenberg (1989) and Boldrin, Kiyotaki, and Wright (1993) find that models with product market slack that are precursors of this book's slackish model exhibit periodic orbits; thus, slackish models might produce endogenous business cycles and may help explain recurring fluctuations in slack. Another way to generate such orbits would be to port the slackish structure to an agent-based setting, where oscillatory or chaotic dynamics can emerge.³

20.2. What determines market price norms?

The second open question is what are the foundations and general expressions for the norms that determine market prices, including market wages.

In most models we have used price norms that were simple rigid functions of the model parameters. We also saw that such rigidity is caused in large part by fairness concerns. It would be fruitful to construct realistic norms for market prices, built to mimic real-world fairness concerns. The concepts of fair wage developed by Akerlof (1980, 1982) and Akerlof and Yellen (1990), as well as the concept of fair price developed by Eyster, Madarasz, and Michailat (2021), might be productive starting points.

The Phillips curve of chapter 14 is built by assuming a quadratic price-adjustment cost coupled with directed search. It provides one direction to endogenize price norms, but it does not yet explicitly incorporate the fairness concerns that seem so important in reality.

Relatedly, it would be invaluable to have more estimates of the passthroughs of marginal valuation into prices, estimates of the passthrough of marginal productivity into wages, and estimates of the passthrough of market tightness into prices. Estimates of these passthroughs would help calibrate price and wage norms realistically.

¹See for instance Akerlof and Romer (1993), Kindleberger and Alibert (2005), Shiller (2005), and Akerlof and Shiller (2015).

²Akerlof and Shiller (2009), Beaudry and Portier (2014), and Nimark (2014) develop these ideas.

³See (Axtell and Farmer 2025) for a survey of the agent-based literature. Gualdi et al. (2015) find such dynamics in a macroeconomic agent-based model with unemployment.

20.3. Why does the Beveridge curve shift?

A third open question is why the Beveridge curve abruptly shifts every one or two decades on the US labor market—as displayed in figure 2.7.

In a dynamic labor market model the location of the Beveridge curve is determined by the job-separation rate—which is observed—and the matching function. So understanding the determinants of the labor market matching function seems important to understanding the shifts of the Beveridge curve, and in turn the movements of the FERU. Typical shifts in the Beveridge curve do not affect the FERU much, but three large shifts over the past century had a sizable impact on the FERU. These three shifts happened during dramatic periods of US history: the Great Depression, World War 2, and the coronavirus pandemic. It is unsurprising that such dramatic events might shift the curve, but it is unclear what the underlying mechanism was.

20.4. Why is the product Okun curve steeper in the model than in the data?

The fourth open question is why the product Okun curve is steeper in the calibrated model than in the data—as apparent by comparing figure 2.12 and figure 15.8D.

One possible explanation for this discrepancy between empirical and model Okun curves is that slack is not well measured on the product market, so the measured idleness rates are more volatile than the true idleness rates. Because the ISM is not a governmental agency, it is not required to be as transparent, so we do not know exactly how their surveys are conducted, how large their sample of firms is, and so on. We saw that the idleness rates are quite volatile—much more than the unemployment rate measured by the BLS—so this could be an explanation. We explained that calibration cannot help solve this issue, but maybe the structure of the two-market model of chapter 15 is not capturing some relevant aspects of the product market.

20.5. Why is the monetary multiplier larger in the model than in the data?

The fifth open question is why the monetary multiplier is larger in the calibrated model than in US data. In the calibrated model, the monetary multiplier sits around 1, between 0.7–2 (figure 13.5), while US estimates of the monetary multiplier sit around 0.5, between 0.1–0.93 (section 16.2.3). When the model is inefficiently slack, the monetary multiplier is above 1, which does not overlap with available estimates.

A possible reason why changes in the federal funds rate are less potent in reality than in the model is that the changes do not affect households directly. Households respond to the interest rates on their credit cards, loans, and mortgages, not directly to the fed funds rate. If these interest rates respond sluggishly or partially to the fed funds rate, then

of course the fed funds rate would have a limited effect on aggregate demand and the economy in reality (Gregor, Melecky, and Melecky 2021).

Another possibility is that the estimates of the monetary multiplier are biased downward due to the difficulty in finding exogenous fluctuations in monetary policy.

20.6. Why is the fiscal multiplier smaller in the model than in the data?

The sixth open question is why the fiscal multiplier seems smaller in the calibrated model than in US data. In the slackish model, fiscal multipliers giving the change in total output from public spending (dY/dG) are necessarily below 1 (equation (13.27)). At the same time, US estimates of such output multipliers are routinely above or well above 1 (Chodorow-Reich 2019), especially in bad times (Auerbach and Gorodnichenko 2012).

A possibility is that the model misses ingredients to amplify the effects of fiscal policy. Candidates include financing of public spending through deficit spending, heterogeneous marginal propensity to consume, complementarities between private and public consumption, and so on. Another possibility is that estimates of the output multiplier above 1 are the exception rather than the norm (Ramey 2019).

This open question need not matter much for the conduct of fiscal policy. In a world with slack, the relevant sufficient statistic is not the effect of public spending on output but the effect on slack—estimates of these multipliers are much more compact and in line with the model prediction. Moreover, in a world with slack, a multiplier above 1 is not required to justify stimulus spending: the government should increase public spending when the economy is too slack as long as the multiplier is positive—which seems uncontroversial.

20.7. What is the welfare cost of unemployment?

The seventh open question is: what is the welfare cost of unemployment relative to employment?

In the book, and especially when we computed the FERU, we assumed based on available evidence that the welfare cost of idleness in unemployment offset the welfare value of home production, so unemployment has zero net welfare value. The assumption was motivated by substantial evidence from sociology, psychology, public health, and medical science that unemployment is detrimental to physical and especially mental health. But few studies have been able to quantify the welfare cost of unemployment. The study that we use, by Borgschulte and Martorell (2018), uses administrative military data to estimate the value of nonwork relative to the value of work. It would be valuable to measure the welfare value of nonwork for other workers and in other datasets to confirm or refine the findings and ascertain more confidently the welfare cost of unemployment.

20.8. What is the welfare cost of recruiting?

The eighth open question is: what is the welfare cost of recruiting?

When we computed the full-employment rate of unemployment (FERU), we saw that a key statistic is the amount of man-hours devoted to recruiting by firms. There is little evidence on that, so we estimated the amount of labor devoted to each vacancy from two distinct surveys. This lack of evidence is not ideal to measure the FERU and unemployment gap in the past, but it could easily be remedied in the future. The BLS would only need to add one question to the Job Opening and Labor Turnover Survey. The recruiting cost could be measured every month by asking firms to report how many man-hours they devote to recruiting in addition to how many vacancies they are currently advertising. As part of the American Time Use Survey, the BLS already asks unemployed workers how many hours they spend every day searching for a job (Aguilar, Hurst, and Karabarbounis 2013). It should be easy to ask firms the same thing: how many man-hours are spent every month recruiting new workers.

20.9. Can aggregate slack be inferred from labor market slack?

The ninth open question is whether labor market slack is a good measure of aggregate slack, especially for policy purposes.

The business cycle models of chapters 13 and 14 use a one-market representation of the economy, blending the labor and product markets. In the spirit of the models, the empirical applications of the models would use a broad measure of aggregate slack, which would account not only for unemployment but also idle labor inside firms, unsold goods, unused capacity, along the lines of the evidence in chapter 2. Constructing such an aggregate slack rate and slack gap is not yet feasible with existing US data, so the empirical applications use the unemployment rate and unemployment gap. But it would be valuable to attempt to construct a measure of aggregate slack and assess whether it yields new insights for assessing aggregate efficiency and conducting monetary policy, compared to the measure of labor market slack used now.

Building measures of aggregate slack and the aggregate slack gap would require collecting more data on product market slack: to know how many goods remain unsold across the economy at a monthly frequency, and how many employees remain idle each month across the economy—waiting for customers to purchase their labor services. To measure the cost of matching—necessary to calculate the efficient slack rate—we would need to know how many buyers there are for goods and services at any point in time, and how much resource is devoted to buying across the economy. Unlike in the labor market, where job vacancies are measured systematically, the desired amount of purchases in the product market is not recorded in any centralized dataset; and there is very little data on

the cost of visiting shops and filling help-wanted ads.

Of course, in the United States, most production is done in firms, so the two-market model of chapter 15 is more realistic. The question in that model is whether we can infer the efficiency of the aggregate economy from labor market slack. The key to answering this question seems to be whether the product-labor coincidence holds or not.

20.10. Does slack also influence longer-run fluctuations?

The tenth and final open question is whether slack only influences short-run fluctuations, or it also affects medium-run and long-run economic fluctuations.

The book's focus has been on short-run issues, namely business cycles. We have not explored at all how economic slack might help us understand longer-run issues, such as economic development and growth. Because of this short-run focus, we have concentrated on labor as a factor of production. We have not paid much attention to capital as a factor of production. We have barely looked at data on capital goods, and we have not explicitly modeled capital and related quantities, such as investment. Yet, capital also remains idle some of the time, and capital utilization varies over time, as we briefly discussed in chapter 2. The slackish framework might be helpful to connect capital utilization to capital accumulation, thus linking slack to growth.

Bibliography

- Aguiar, Mark, Erik Hurst, and Loukas Karabarbounis. 2013. "Time Use during the Great Recession." *American Economic Review* 103 (5): 1664–1696. <https://doi.org/10.1257/aer.103.5.1664>.
- Akerlof, George A. 1980. "A Theory of Social Custom, of Which Unemployment May Be One Consequence." *Quarterly Journal of Economics* 94 (4): 749–775. <https://doi.org/10.2307/1885667>.
- Akerlof, George A. 1982. "Labor Contracts as Partial Gift Exchange." *Quarterly Journal of Economics* 97 (4): 543–569. <https://doi.org/10.2307/1885099>.
- Akerlof, George A. and Paul M. Romer. 1993. "Looting: The Economic Underworld of Bankruptcy for Profit." *Brookings Papers on Economic Activity* 1993 (2): 1–73. <https://doi.org/10.2307/2534564>.
- Akerlof, George A. and Robert J. Shiller. 2009. *Animal Spirits: How Human Psychology Drives the Economy, and Why It Matters for Global Capitalism*. Princeton, NJ: Princeton University Press.
- Akerlof, George A. and Robert J. Shiller. 2015. *Phishing for Phools: The Economics of Manipulation and Deception*. Princeton, NJ: Princeton University Press.
- Akerlof, George A. and Janet L. Yellen. 1990. "The Fair Wage-Effort Hypothesis and Unemployment." *Quarterly Journal of Economics* 105 (2): 255–283. <https://doi.org/10.2307/2937787>.
- Auerbach, Alan J. and Yuriy Gorodnichenko. 2012. "Measuring the Output Responses to Fiscal Policy." *American Economic Journal: Economic Policy* 4 (2): 1–27. <https://doi.org/10.1257/pol.4.2.1>.
- Axtell, Robert L. and J. Doyne Farmer. 2025. "Agent-Based Modeling in Economics and Finance: Past, Present, and Future." *Journal of Economic Literature* 63 (1): 197–287. <https://doi.org/10.1257/jel.20221319>.
- Beaudry, Paul, Dana Galizia, and Franck Portier. 2020. "Putting the Cycle Back into Business Cycle Analysis." *American Economic Review* 110 (1): 1–47. <https://doi.org/10.1257/aer.20190789>.
- Beaudry, Paul and Franck Portier. 2014. "News-Driven Business Cycles: Insights and Challenges." *Journal of Economic Literature* 52 (4): 993–1074. <https://doi.org/10.1257/jel.52.4.993>.
- Boldrin, Michele, Nobuhiro Kiyotaki, and Randall Wright. 1993. "A Dynamic Equilibrium Model of

- Search, Production, and Exchange.” *Journal of Economic Dynamics and Control* 17 (5–6): 723–758. [https://doi.org/10.1016/0165-1889\(93\)90012-H](https://doi.org/10.1016/0165-1889(93)90012-H).
- Borgschulte, Mark and Paco Martorell. 2018. “Paying to Avoid Recession: Using Reenlistment to Estimate the Cost of Unemployment.” *American Economic Journal: Applied Economics* 10 (3): 101–127. <https://doi.org/10.1257/app.20160257>.
- Chodorow-Reich, Gabriel. 2019. “Geographic Cross-Sectional Fiscal Spending Multipliers: What Have We Learned?” *American Economic Journal: Economic Policy* 11 (2): 1–34. <https://doi.org/10.1257/pol.20160465>.
- Diamond, Peter and Drew Fudenberg. 1989. “Rational Expectations Business Cycles in Search Equilibrium.” *Journal of Political Economy* 97 (3): 606–619. <https://doi.org/10.1086/261618>.
- Eyster, Erik, Kristof Madarasz, and Pascal Michailat. 2021. “Pricing under Fairness Concerns.” *Journal of the European Economic Association* 19 (3): 1853–1898. <https://doi.org/10.1093/jeea/jvaa041>.
- Gregor, Jiri, Ales Melecky, and Martin Melecky. 2021. “Interest Rate Pass-Through: A Meta-Analysis of the Literature.” *Journal of Economic Surveys* 35 (1): 141–191. <https://doi.org/10.1111/joes.12393>.
- Gualdi, Stanislao, Marco Tarzia, Francesco Zamponi, and Jean-Philippe Bouchaud. 2015. “Tipping Points in Macroeconomic Agent-Based Models.” *Journal of Economic Dynamics and Control* 50: 29–61. <https://doi.org/10.1016/j.jedc.2014.08.003>.
- Keynes, John Maynard. 1936. *The General Theory of Employment, Interest and Money*. New York: Macmillan.
- Kindleberger, Charles P. and Robert Z. Alibert. 2005. *Manias, Panics and Crashes: A History of Financial Crises*. 5th ed. New York: Palgrave Macmillan.
- Nimark, Kristoffer P. 2014. “Man-Bites-Dog Business Cycles.” *American Economic Review* 104 (8): 2320–2367. <https://doi.org/10.1257/aer.104.8.2320>.
- Ramey, Valerie A. 2019. “Ten Years after the Financial Crisis: What Have We Learned from the Renaissance in Fiscal Research?” *Journal of Economic Perspectives* 33 (2): 89–114. <https://doi.org/10.1257/jep.33.2.89>.
- Shiller, Robert J. 2005. *Irrational Exuberance*. 2nd ed. Princeton, NJ: Princeton University Press.